- Suppose the strategy set of player 1 is $S_1 = \{s_{11}, s_{12}, s_{13}, ...s_{1J}\}$. Player 1 has J actions or strategies. Strategy set of player 2 is $S_1 = \{s_{21}, s_{22}, s_{23}, ...s_{2K}\}$, player 2 has K strategies or actions.
- If player 1 believes that player 2 will play the strategies $(s_{21}, s_{21}, ...s_{2k})$ with the probabilities $(p_{21}, p_{22}, ..., p_{2k})$ then player 1's expected payoff from playing s_{1j} is $\sum_{k=1}^{K} p_{2k} u_1(s_{1j}, s_{2k})$.
- The expected payoff of player 1 from playing mixed strategy $(p_{11}, p_{12}, p_{13}, , , , p_{1J})$ is $E_1(p_i, p_2) = \sum_{j=1}^{J} p_{1j} \left[\sum_{k=1}^{K} p_{2k} u_1(s_{1j}, s_{2k}) \right] = \sum_{j=1}^{J} \sum_{k=1}^{K} p_{1j} p_{2k} u_1(s_{1j}, s_{2k})$.

- Similarly the expected payoff of player 2 from playing mixed strategy $(p_{21}, p_{22}, p_{23}, , , , p_{2K})$ is
$E_2(p_i, p_2) = \sum_{k=1}^{K} p_{2k}\left[\sum_{j=1}^{J} p_{1j}u_2(s_{1j}, s_{2k})\right] = \sum_{k=1}^{K}\sum_{j=1}^{J} p_{1j}p_{2k}u_2(s_{1j}, s_{2k})$.
- Mixed Strategy Nash equilibrium
In a two player normal form game $G = \{S_1, S_2; u_1, u_2\}$, the mixed strategies (p_1^*, p_2^*) are a Nash equilibrium if each player's mixed strategy is a best response to the other player's mixed strategy that is the two conditions given below must hold
$E_1(p_1^*, p_2^*) \geq E_1(p_1, p_2^*)$ for every probability distribution p_1 over S_1.
$E_2(p_1^*, p_2^*) \geq E_2(p_1^*, p_2)$ for every probability distribution p_2 over S_2.